23145

C23-M–**304**

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – THIRD SEMESTER EXAMINATION

STRENGTH OF MATERIALS

Time : 3 Hours] [*Total Marks* : 80

PART—A 3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define the following terms :

(a) Poisson's ratio

(b) Modulus of elasticity

(c) Modulus of rigidity

2. Draw the stress-strain diagram for ductile material and locate salient points on it.

3. Define factor of safety. What is the significance of factor of safety?

4. Define *(a)* Resilience, *(b)* Proof Resilience and *(c)* Modulus of Resilience.

5. List any three types of loads that are applied on various beams.

6. Draw the shear force diagram for a cantilever beam subjected to a point load at its free end.

7. State any three assumptions made in theory of simple bending.

8. A cantilever beam of length 6m is carrying a uniform distributed load of 16 kN-m. Calculate the deflection at the free end of the beam. Take moment of inertia = 95×10^7 mm^4 and $E = 2 \times 10^5$ N/mm2.

9. Write the Torsion equation and mention the units of the terms.

10. Define the terms spring index and stiffness related to coiled helical springs

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

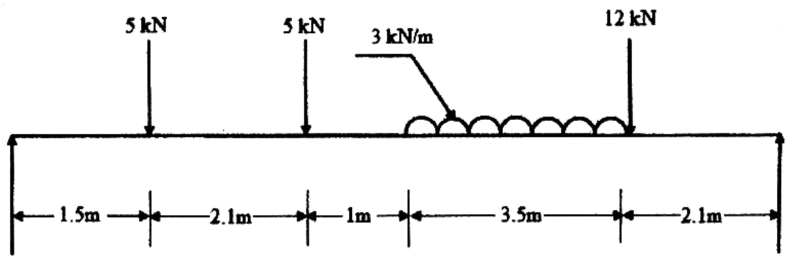
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. A steel bar 600 mm long is 28 mm in diameter for 240 mm of its length, 20 mm in diameter for 200 mm of its length and 16 mm in diameter for the remaining length. It is subjected to an axial pull of 20 kN. If $E = 2.1 \times 10^5$ N/mm^2, calculate the extension of the bar and stress in each section.

12. An MS bar of length 2 m has a diameter of 50 mm, hangs vertically. A load of 20 kN falls on a collar attached to the lower end. Find the maximum stress when *(a)* height of fall is 100 mm, *(b)* load is applied suddenly without impact and *(c)* load is applied gradually. Take $E = 2.1 \times 10^5$ N/mm^2.

13. A 6 m long cantilever beam carries loads of 2 kN and 3 kN at 2 m and 5 m respectively from fixed end. Draw S.F and B.M diagram for the beam.

14. Draw SF and BM diagrams for the beam loaded as shown in figure below. All loads are in kN and length are in metre :



15. A rectangular beam 200 mm deep and 300 mm wide is simply supported over a span of 8 m. If the bending stress is not to exceed 120 N/mm^2, what is the uniformly distributed load per metre required?

16. *(a)* A timber beam, 150 mm × 300 mm cross section supports a central point load on a span of 4 m. If the maximum bending stress is 8 N/mm2, what is the maximum deflection?

Take $E = 0.1 \times 10^5$ N/mm^2.

(b) A cantilever 1.25 m long of section 100 mm wide × 160 mm deep carries a concentrated load of 60 kN at free end. Find the deflection at free end. Take $E = 2.1 \times 10^5$ N/mm^2.

17. A hollow circular shaft external diameter 50 mm and internal diameter 40 mm transmit a torque of 40 kN-m. Find the maximum shear stress induced in the shaft.

18. A helical, in which the slope of the helix may be assumed to be negligible, is required to transmit a maximum pull of 1 kN and to extend 10 mm for 200 N load. If the mean diameter of the coil is 80 mm, find the suitable diameter for the wire and number of coils required. Take G = 80 GPa and allowable shear stress as 100 MPa.

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